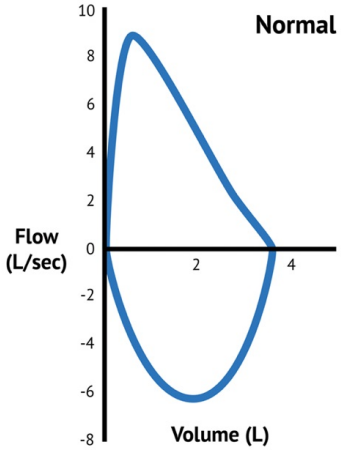
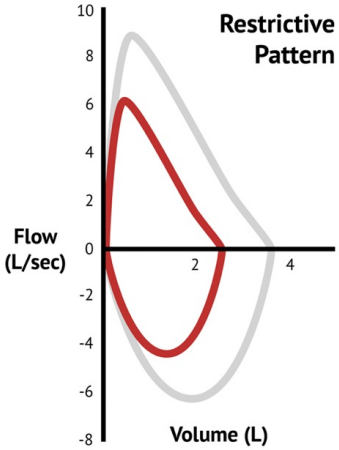
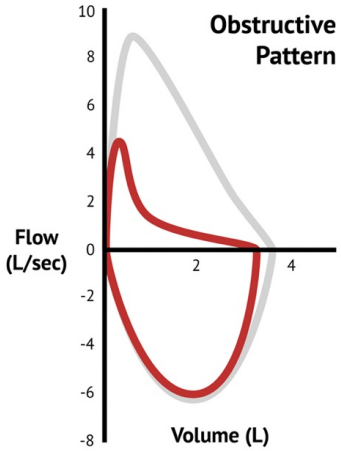
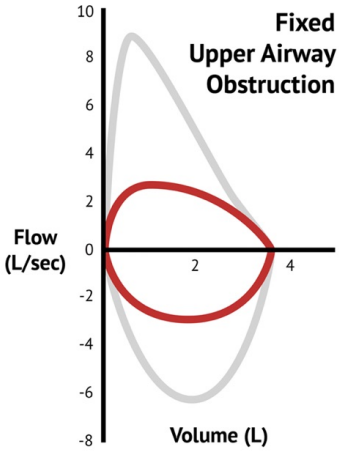
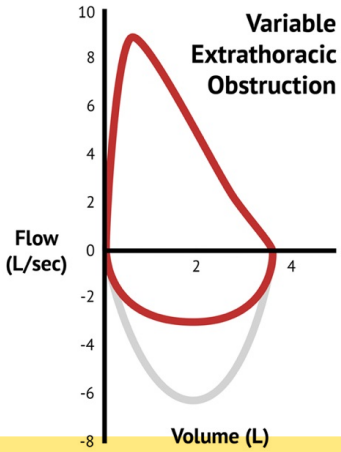
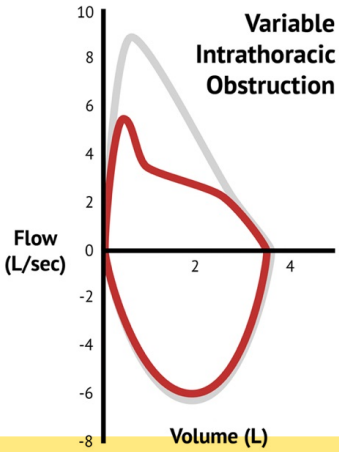






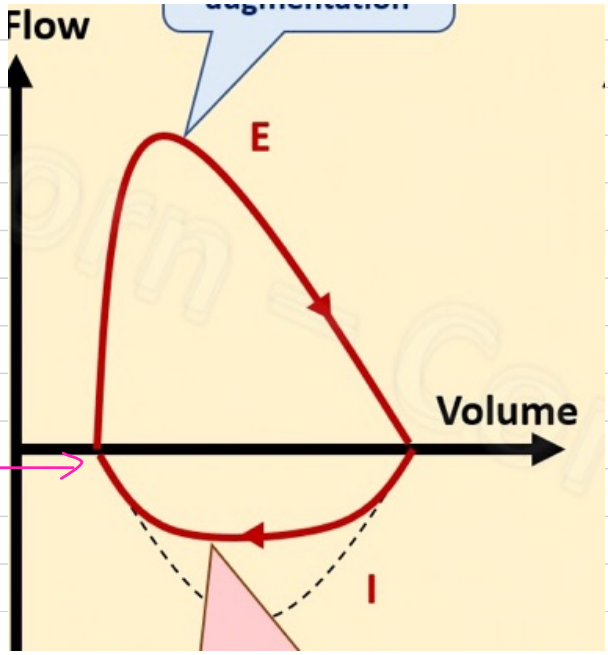






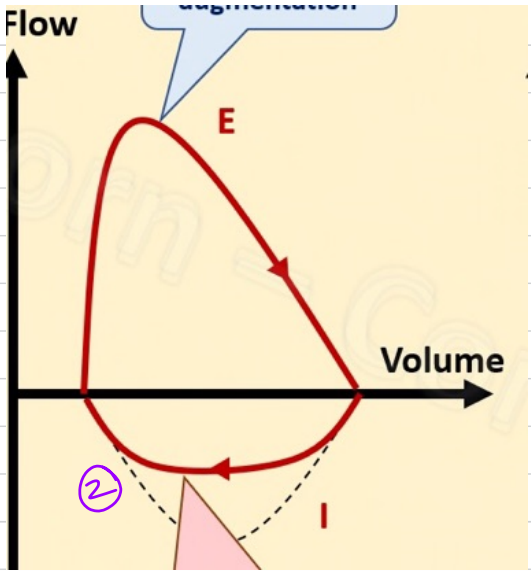
A 60yr old chronic smoker presented to emergency with acute onset dyspnea and an inspiratory stridor. No associated dysphagia present. On examination the body temperature was normal and the chest examination also came out to be normal. Local oral cavity and oropharynx examination was normal. Laryngoscopy could not be done. A previous PFT was done for the patient which showed the following type of flow volume curve.

What is the probable site of lesion and what can be the cause?



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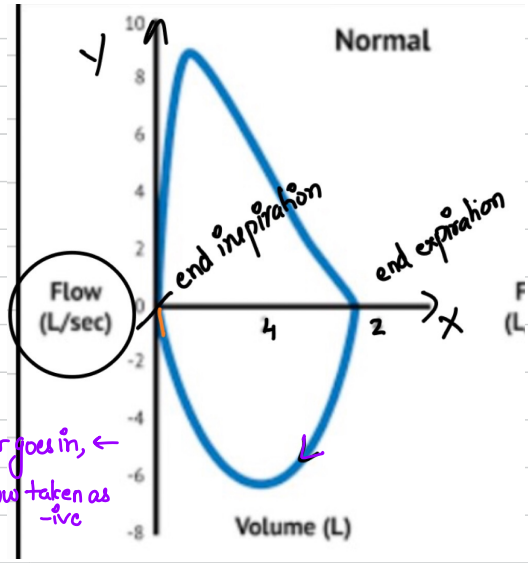
What is the probable site of lesion and what can be the cause?



1st step → look for the phase which is affected?

↓

Inspiratory vs Expiratory vs Biphasic



① & ② → Inspiratory phase affected

Anatomical location based on the phase affected

- Inspiratory → Pharynx or Supraglottis; Vocal cord
- Biphasic → Vocal cord; Subglottis, Tracheal, Extrathoracic airway
- Expiratory → Intrathoracic airway

BASIC PHYSICS LAWS → ① Volume ↑ → Pressure ↓ (Boyle's law)

② Air moves from high (P) → low (P)

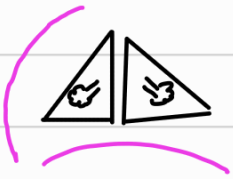


(N) respiration cycle → Initially active inspiration f/b passive expiration

Pre-Botzinger

Inspiratory centre (DRG) (2s)
- → ⊕ AHC of C_3, C_4, C_5 motor roots → Diaphragm contract
- → ⊕ AHC of Thoracic motor roots → ICs contract

→ Intrathoracic volume ↑



↓

Intrathoracic pressure becomes -ive

↓

Pulls lung outward → Airways (in long parenchyma open → Airway volume ↑ sec → Pressure (in airway becomes less (-ive)

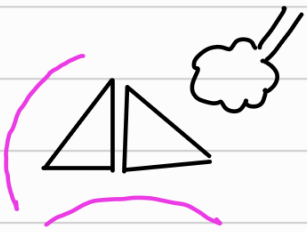
↓

Air (P) > Airway (P) → Air goes into airways

When distension of alveoli is adequate

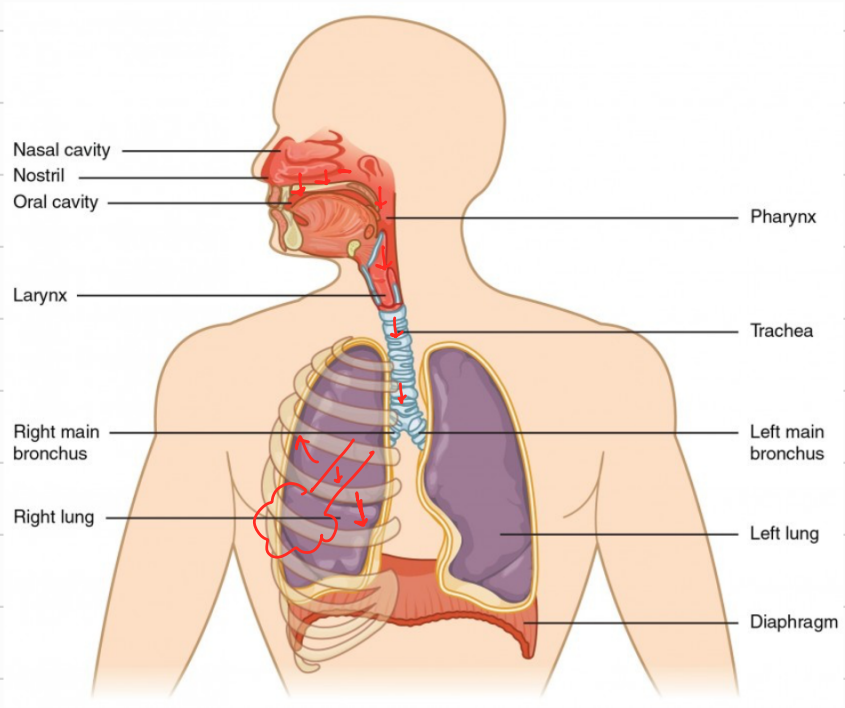
⊕ of vagus → ⊕ Pneumotaxic → ⊖ Apneustic

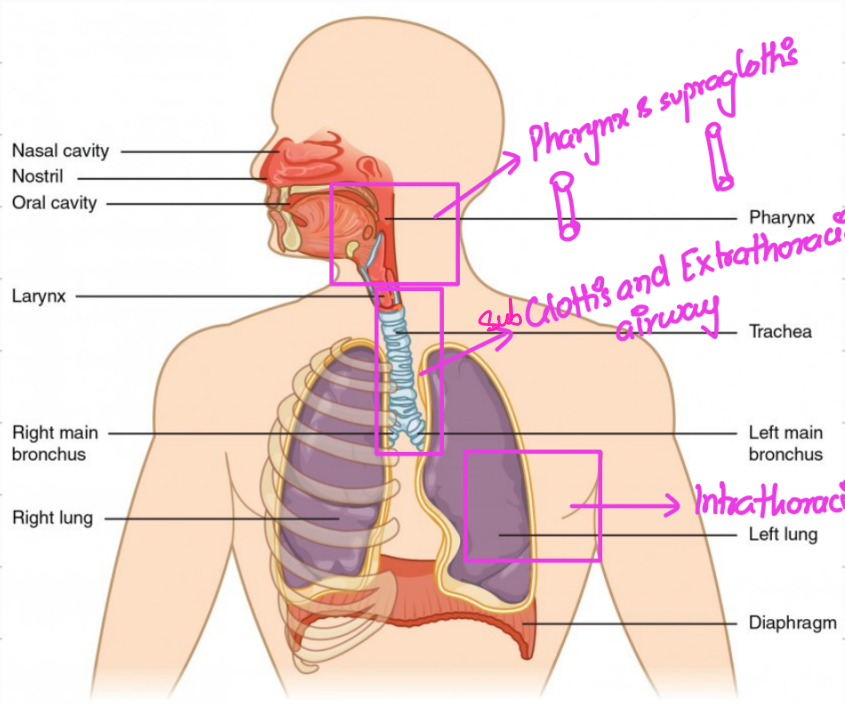
⊕ DRG ↗ no ⊕ion of muscles → Chest goes in → Intrathoracic pressure ↑



↓

Lung airway forced to close → Volume ↓ → Pressure ↑ → Airway (P) > Air (P) → Air out





on inspiration, -ive pressure created in air way to suck in air through nose





Supraglottis

membrane collapse, calibre during inspiration



more turbulent during inspiration

Extrathoracic variable

Cricoid & Extrathoracic



Hard cartilage ⊕

Doesn't let airways collapse

Same calibre in inspiration & expiration

Turbulence d/t obstruction same in (I) & (E)

Extrathoracic fixed

Intrathoracic

on inspiration elastic fibres stretch airway

calibre ↑ on inspiration

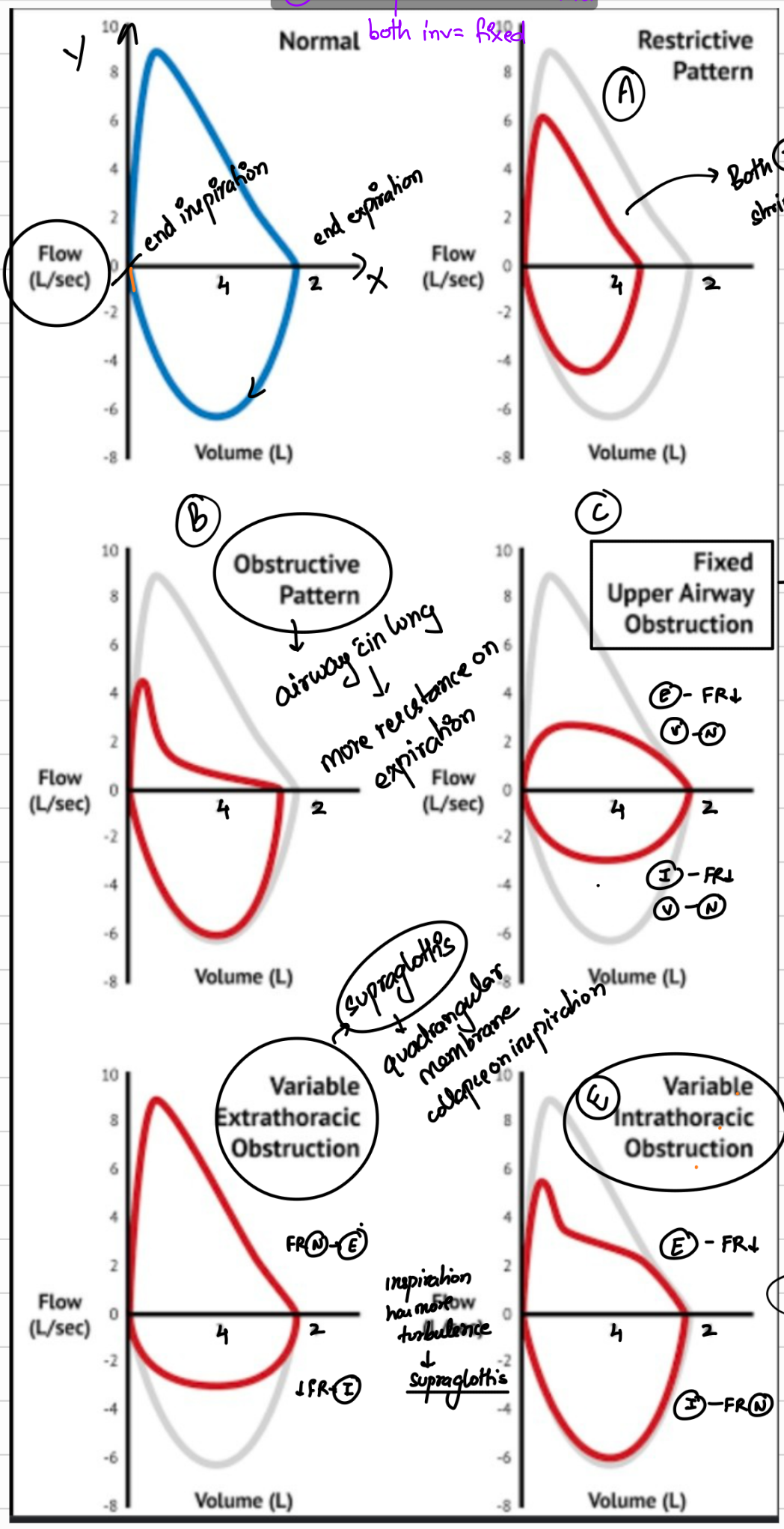
Turbulence more in expiration

same reason why wheez in expir

Intrathoracic Variable

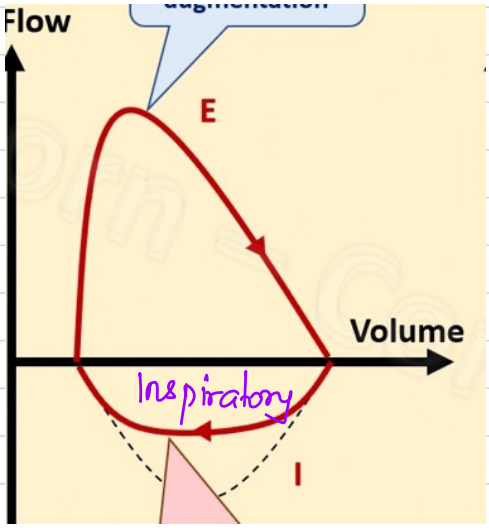
① Check the phase involved
② If one phase inv = Variable
both inv = fixed





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What is the probable site of lesion and what can be the cause?



Inspiratory → Pharynx vs Supraglottis

↓

oral cavity OR oropharynx exⁿ → (N)

Mass in supraglottis + smoker + No fever → ? Ca larynx

Lesion in hypopharynx compressing the supraglottis → No dysphagia (+)

Acute → ? Hemmorrhage

↓

Direct Laryngoscopy